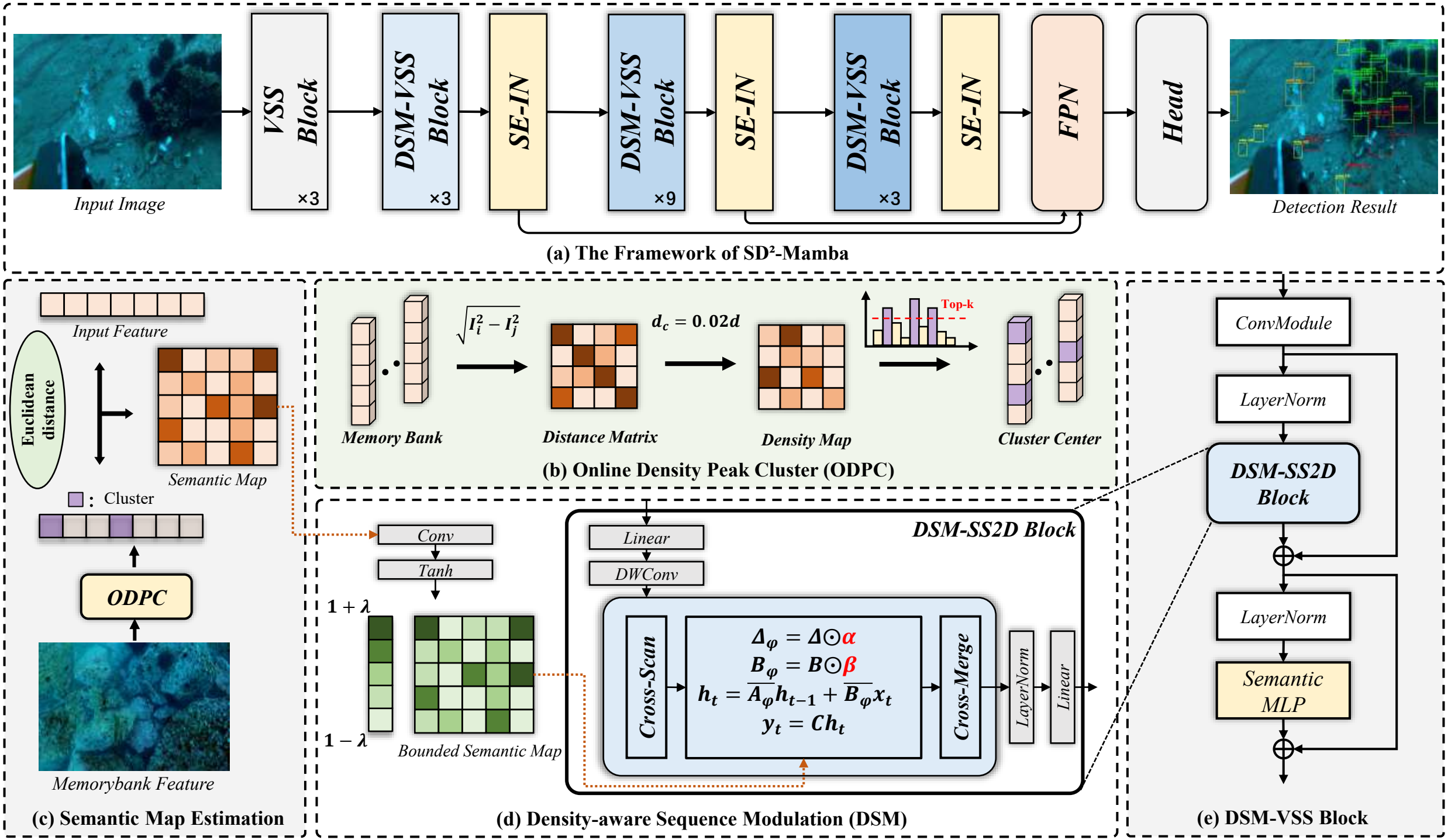
Input Image
VSS Block ×3
DSM-VSS Block ×3
SE-IN
DSM-VSS Block ×9
SE-IN
DSM-VSS Block ×3
SE-IN
FPN
Head
Detection Result
(a) The Framework of SD²-Mamba
Input Feature
Euclidean distance
Semantic Map
: Cluster
ODPC
Memorybank Feature
(c) Semantic Map Estimation
Memory Bank
Distance Matrix
$d_c = 0.02d$
Density Map
Top-k
Cluster Center
(b) Online Density Peak Cluster (ODPC)
Conv
Tanh
$1+\lambda$
$1-\lambda$
Bounded Semantic Map
DSM-SS2D Block
Linear
DWConv
Cross-Scan
$\Delta_\varphi = \Delta \odot \alpha$
$B_\varphi = B \odot \beta$
$h_t = \overline{A_\varphi} h_{t-1} + \overline{B_\varphi} x_t$
$y_t = C h_t$
Cross-Merge
LayerNorm
Linear
(d) Density-aware Sequence Modulation (DSM)
ConvModule
LayerNorm
DSM-SS2D Block
LayerNorm
Semantic MLP
(e) DSM-VSS Block